Hasan Mohd Hussain

Machine Learning Engineer | *Agentic AI*
Mumbai | 7400341331 | hasan98209@gmail.com | Portfolio | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Machine Learning Engineer with an **MS in Computer Engineering**, building **Generative AI**, **RAG** and **agentic AI systems**. Shipped **LLM evaluation and test automation** that reduced QA time by **95%**. Experience in **PyTorch**, **LangChain/LangGraph**, **vector search**, **model deployment**, **FastAPI** and **Docker**.

EDUCATION

University of Texas at Dallas — **Richardson, Texas, USA**
Master of Science in Computer Engineering — Aug 2024 - Aug 2026
GPA: 3.74/4.0 | *Relevant Coursework: NLP, Machine Learning, Computer Architecture, Embedded Systems*

University of Mumbai — **Mumbai, Maharashtra, India**
Bachelor of Engineering in Computer Engineering — Aug 2017 - Jul 2021
GPA: 7.88/10.0 | *Relevant Coursework: Data Structures, Database Management Systems*

SKILLS

Machine Learning: Supervised learning, deep learning, NLP, computer vision, CNNs, transformers, model evaluation
Generative AI: LLMs, RAG, embeddings, semantic search, prompt engineering, LLM-as-a-judge, tool calling
Agentic Systems: LangChain, LangGraph, ReAct, multi-agent orchestration, Critic-Revise loops, streaming, Claude Cowork
Frameworks & APIs: PyTorch, scikit-learn, Hugging Face, Pandas, NumPy, OpenAI API, Azure OpenAI, Gemini API, Ollama, Claude Code
Deployment & Data: FastAPI, Flask, Docker, AWS, CI/CD, ChromaDB, PostgreSQL, MongoDB, SQLite
Languages & Quality: Python, SQL, Java, C++, Git, Linux, PyTest, Playwright, validation-based hyperparameter tuning

WORK EXPERIENCE

Mphasis — **New York, United States**
Machine Learning Engineer — May 2025 - Aug 2025, Jun 2026 - Aug 2026
- Developed an **autonomous LLM QA agent** using **Python, Flask and Playwright** for a production mortgage dashboard.
- Applied **LLM-as-a-judge evaluation** across **Claude, GPT-4o, Gemini and Qwen**, cutting per-case test time by **95%**.
- Generated **synthetic financial documents** and answer keys across **5 document types**, enabling **200 evaluation cases per run**.
- Built a **RAG ingestion and retrieval pipeline** with **ChromaDB, FastAPI and Azure OpenAI**, improving retrieval by **30%**.
- Optimized **Python backend logic** for full-stack AI services, improving **API response time by 20%**.

Dot9 Games — **Mumbai, India**
Game Developer Intern — Apr 2024 - Aug 2024
- Built gameplay systems in **Unity** and **C#** from design documents, shipping features that raised **user engagement by 25%**.
- Ran **POC testing** and **QA** on new game designs and optimized performance to a stable **60 FPS**, managing code with **Git**.

Shree Infotech — **Mumbai, India**
Software Engineer — Apr 2023 - Mar 2024
- Developed **Python/Flask backend services** and **REST APIs** on **Docker**, automating workflows and saving **10+ hours/week**.
- Designed **MongoDB** schemas and indexes for **fast query retrieval**, and wrote **unit tests** (PyTest) to keep core services reliable.

Nilee Games — **Mumbai, India**
Game Developer Intern — Nov 2022 - Mar 2023
- Built a **2D physics-based driving game** in **Unity/C#** within **two weeks**, contributing to **500K downloads** in its first month.
- Raised **user engagement by 30%** through gameplay tuning and **QA testing** on Android devices, using **Git** for source control.

PROJECTS

NutriCoach.AI | *LangGraph, LangChain, GPT-4o-mini, Flask SSE*
- Built **multi-agent orchestration** with **ReAct tool calling** and a **Critic-Revise feedback loop** in **LangGraph**.
- Streamed **Think-Act-Observe traces** through **Flask SSE** to deliver **7-day meal plans** and **90-day roadmaps**.

RAG Document QA System | *LangChain, ChromaDB, Gemini*
- Built **PDF ingestion, text chunking and embedding pipelines** using **Hugging Face** and **ChromaDB semantic search**.
- Integrated **Gemini** for **context-grounded answers** with **source metadata** and retrieved document passages.

Deep Learning Image Classification | *PyTorch, CNN, MLP*
- Trained **CNN/MLP architectures** with **BatchNorm, Dropout** and **30+ hyperparameter experiments** on MNIST/CIFAR-10.
- Achieved **99%+ MNIST accuracy** and **78% CIFAR-10 accuracy**; improved on the **MLP baseline by 25+ points**.

CERTIFICATIONS

Anthropic: Claude Code in Action, Claude Code 101, Claude 101 | **Udemy:** The Complete SQL Bootcamp
IBM: Machine Learning with Python, Building AI Agents and Agentic Workflows | **DeepLearning.AI:** Supervised ML, AI For Everyone